FROM BIAS TO FAIRNESS

CAN SYNTHETIC DATA BRIDGE THE GAP IN TABULAR MACHINE LEARNING?

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- @mlab.ufrgs

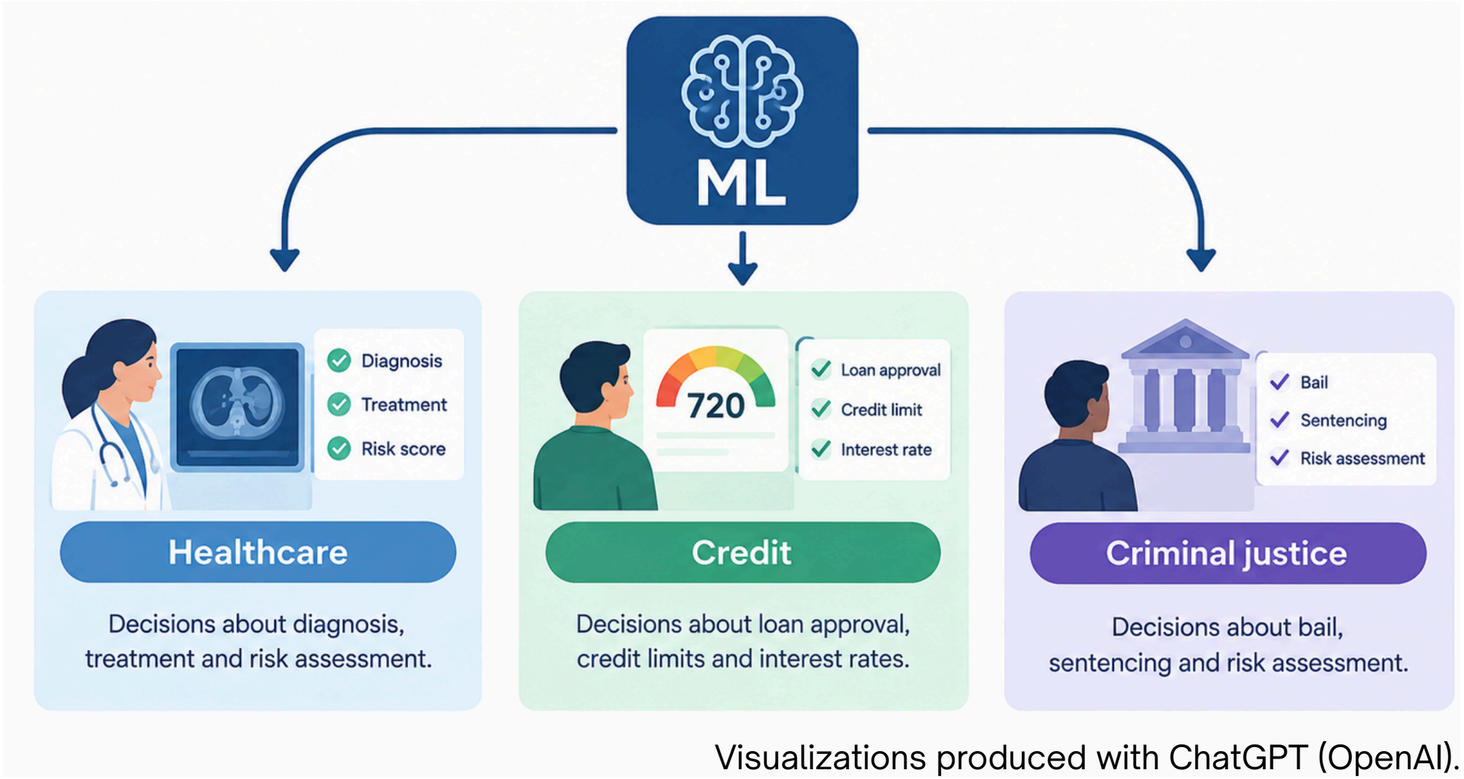
Funded by: CAPES | FAPERGS [22/2551-0000390-7] | CNPq [308075/2021-8]

MOTIVATION [Mehrabi et al., 2021; Caton & Haas, 2024]

Biased data, unequal decisions

1 High-stakes deployment

ML now mediates decisions in healthcare, credit and criminal justice.



Visualizations produced with ChatGPT (OpenAI).

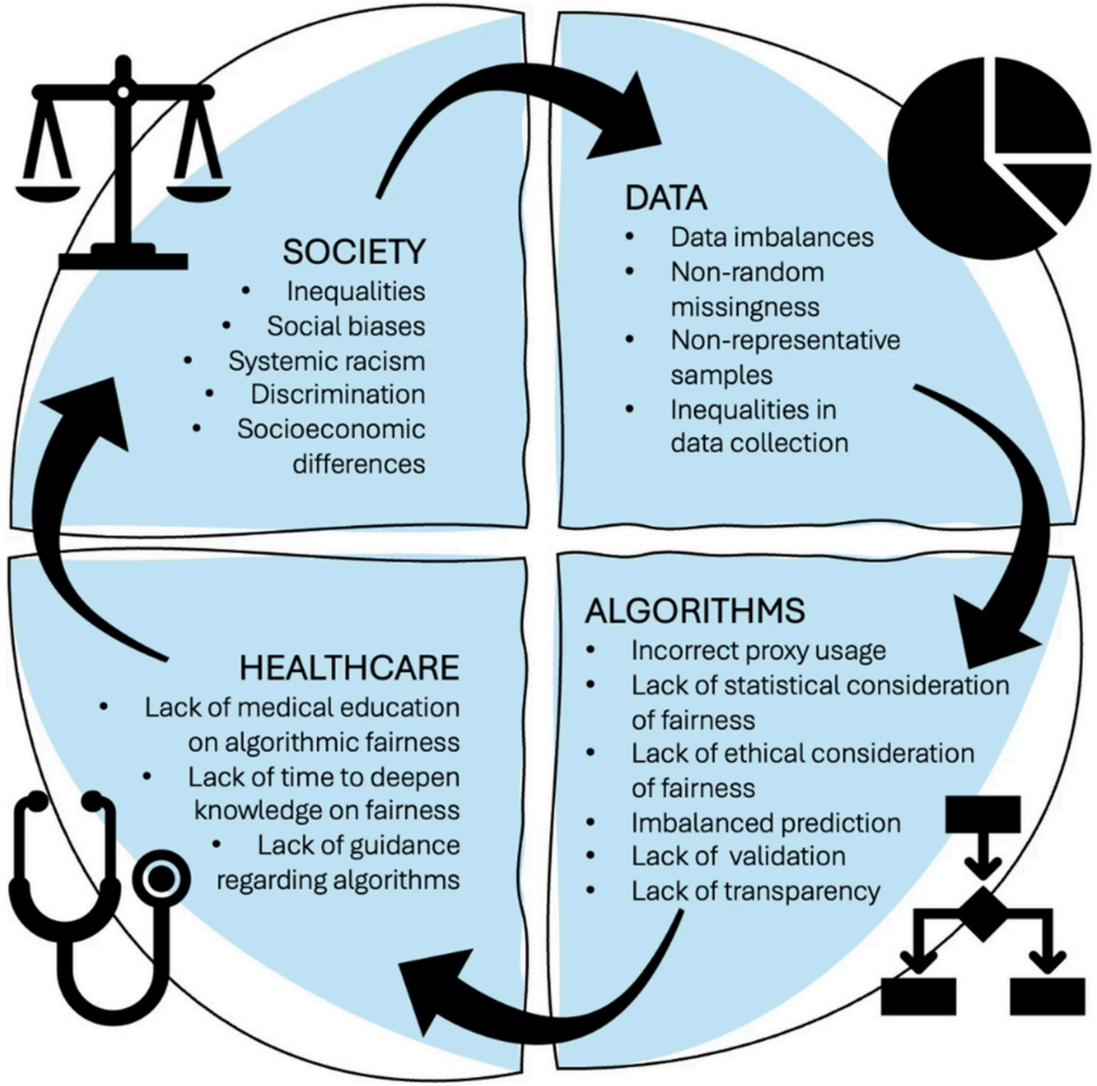
MOTIVATION [Mehrabi et al., 2021; Caton & Haas, 2024]

Biased data, unequal decisions

2

The bias lives in the data

Imbalanced, non-representative samples and structural inequality not only model choice.



Source: Figure 2, extracted from Chakradeo et al., Navigating Fairness Aspects of Clinical Prediction Models.

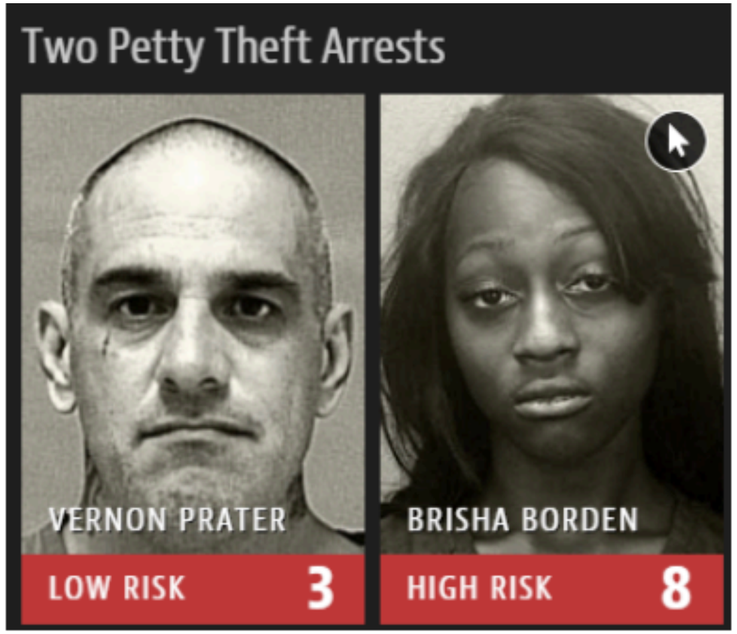
MOTIVATION [Mehrabi et al., 2021; Caton & Haas, 2024]

Biased data, unequal decisions

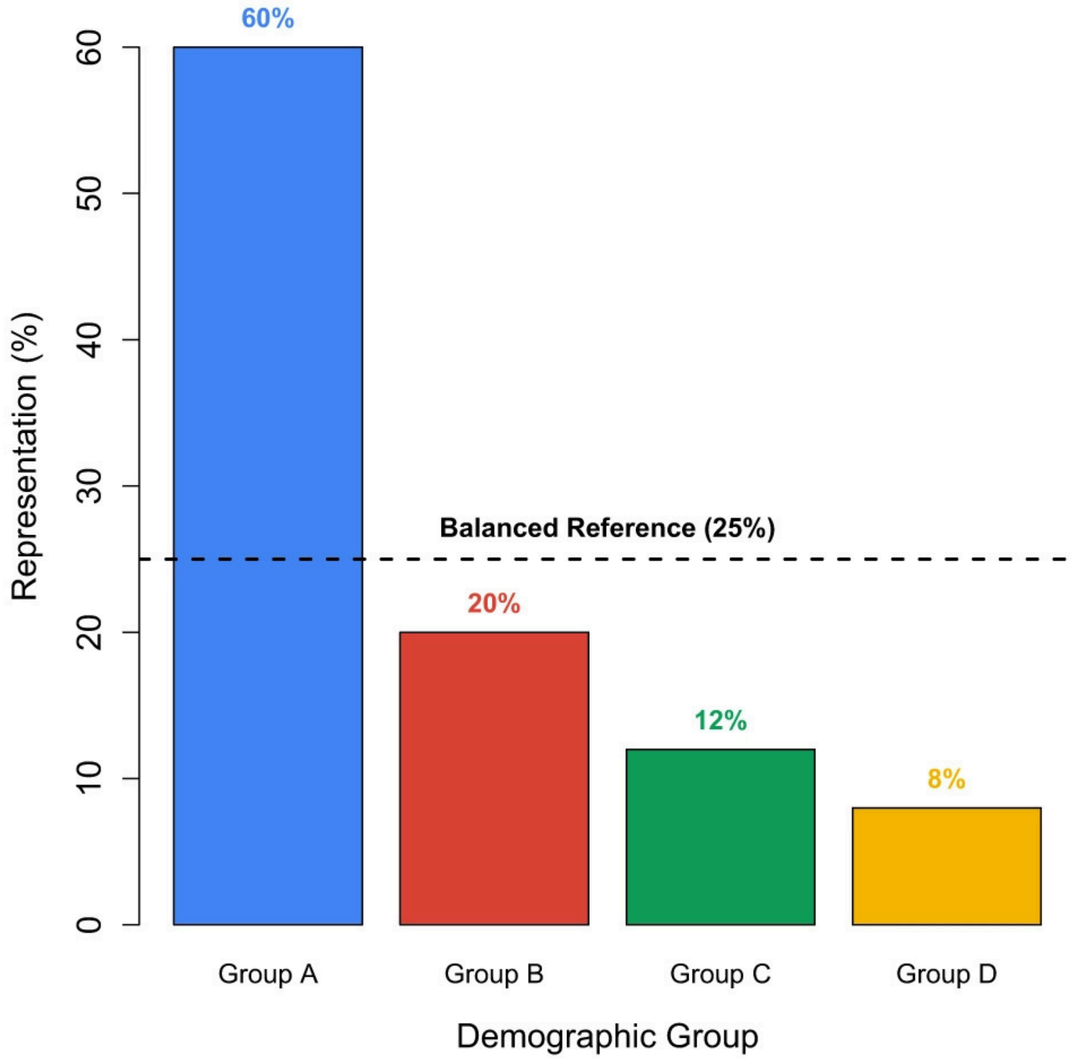
3

The consequence

Systematically different outcomes across demographic groups, and limited safe deployment.



Example extracted from the following source: Fairness and Ethics in Artificial Intelligence, by Aditya.

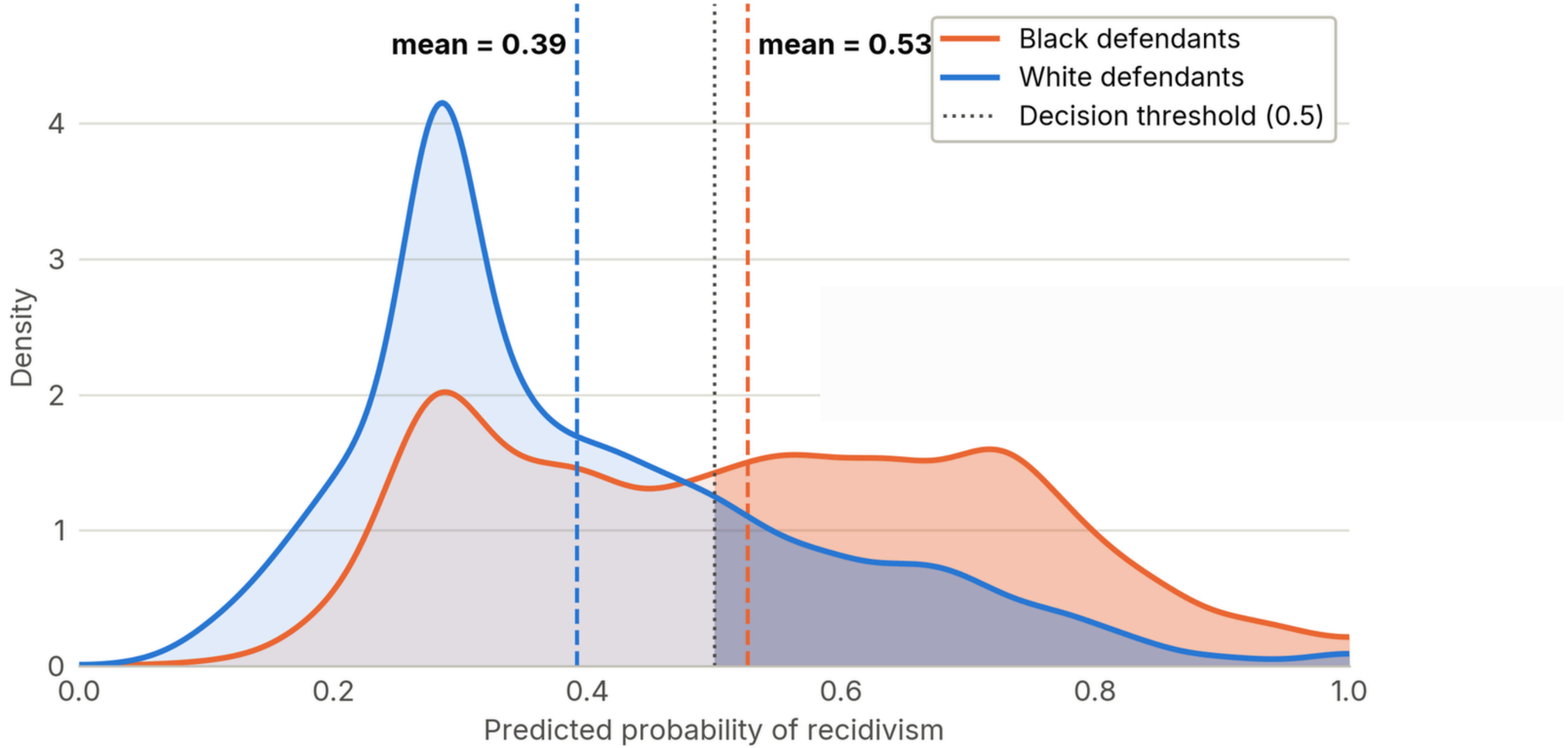


Source: Figure 4, extracted from Ahmad, Vallès, and Idaghdour, Bias in AI Systems: Integrating Formal and Socio-Technical Approaches.

MOTIVATION [Mehrabi et al., 2021; Caton & Haas, 2024]

Compas

African-American defendants are flagged high-risk at more than twice the rate of Caucasian defendants.

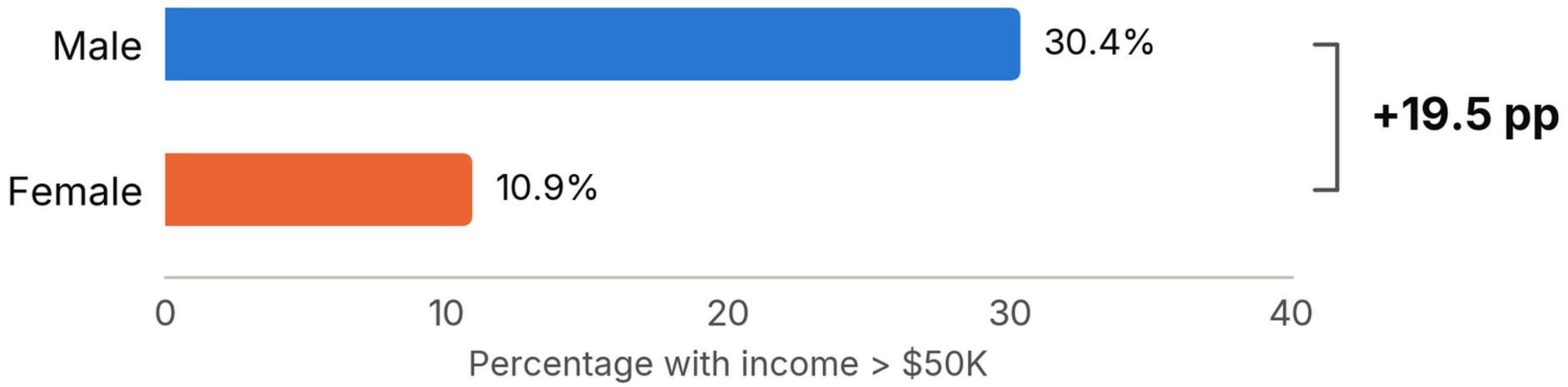


Produced by the authors.

MOTIVATION [Mehrabi et al., 2021; Caton & Haas, 2024]

Adult Income

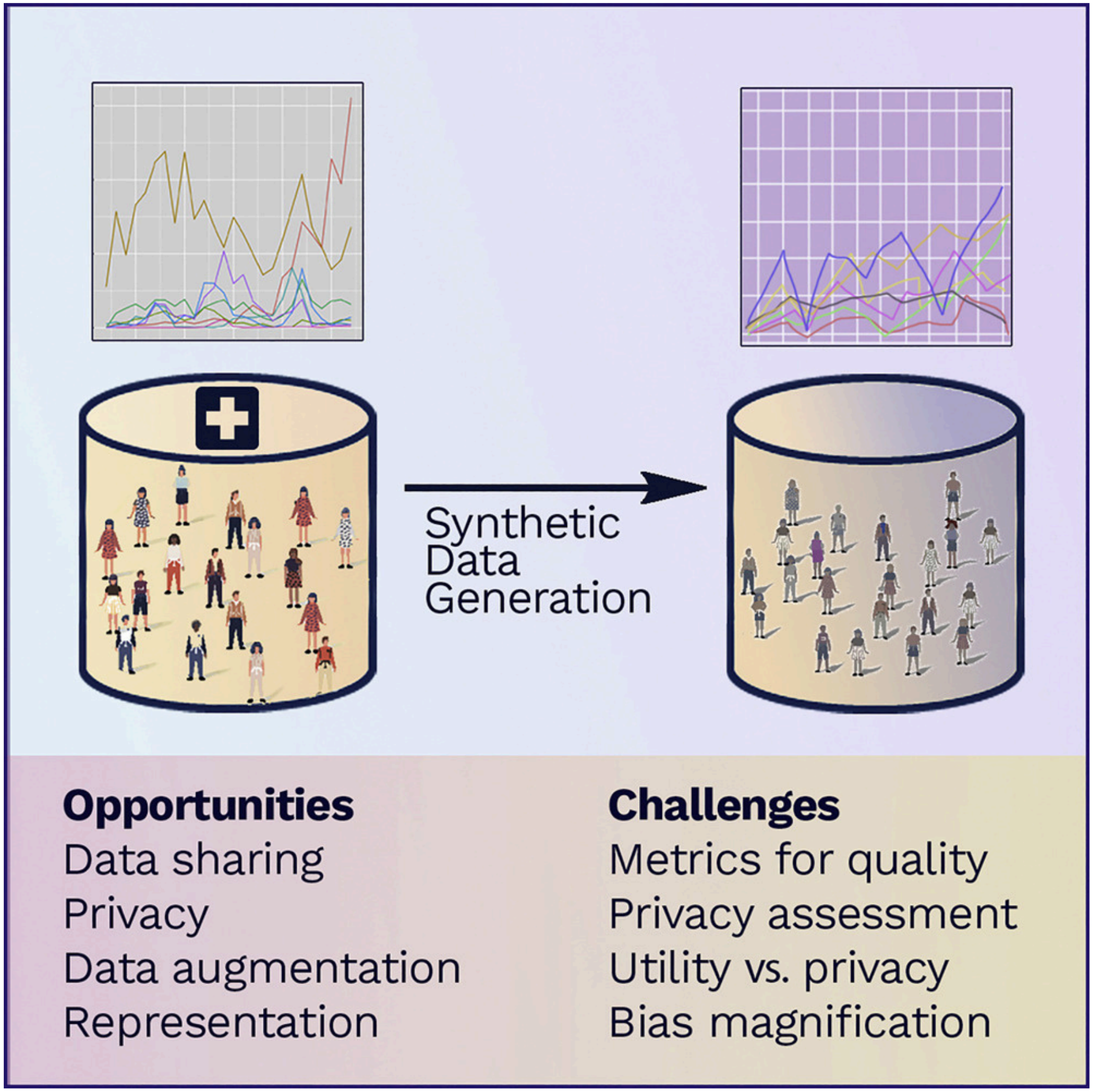
Women receive the favorable (>$50K) prediction at roughly one third the rate of men.



Produced by the authors.

BACKGROUND

Synthetic data



Source: Figure extracted from Rajotte et al., Synthetic Data as an Enabler for Machine Learning Applications in Medicine.

RESEARCH GAP

What prior work leaves open

- **One generator at a time**
- **One fixed pipeline**
- **One fairness metric**

OUR QUESTION

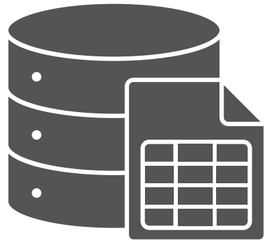
When and how does synthetic data help reduce, or propagate, bias in classification models trained on tabular data?

Related work: [Panagiotou et al., 2024; Liu et al., 2025; Hort et al., 2024.]

STUDY DESIGN

Datasets, preprocessing

1 Datasets



- COMPAS
- Adult
- Diabetes

2 Data preprocessing

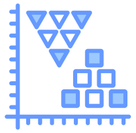
Standard steps



Feature selection



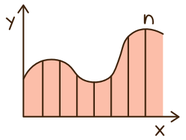
Define protected attribute (A)



Encode categorical



Train / Test split (80/20)



Normalize numerical

Dataset	#Instances	#Attributes	Target	Protected Attribute	Favorable Class
COMPAS	5,278	11	two_year_recid	race	No recidivism (0)
Adult Income	10,000	55	income	sex	>50K (1)
Diabetes	10,000	40	readmitted	race	No readmission (0)

STUDY DESIGN

3. Bias Mitigation / Data Generation

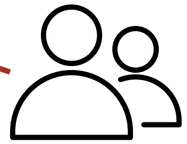
Pre-processing Techniques

Reduces bias by adjusting example weights during training.



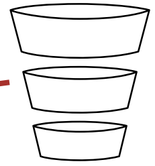
Reweighing

Transforms data attributes to reduce bias between groups.



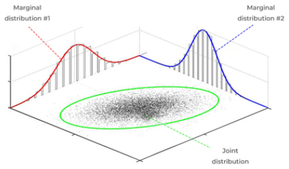
DIR

It learns data representations that reduce bias.



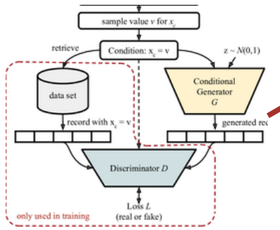
LFR

Synthetic Data Generators (SDG)



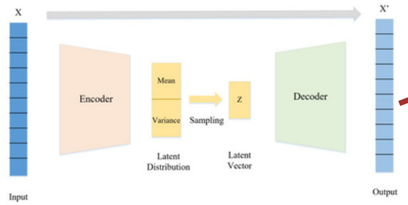
Copula

Preserving distributions and dependencies between variables



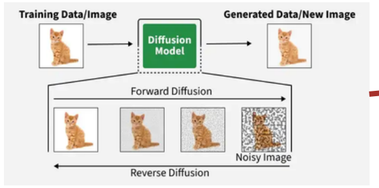
Generative adversarial networks

Conditional Adversarial Generative Network



Autoenconder

Learns a compact probabilistic representation of the data.



Diffusion model

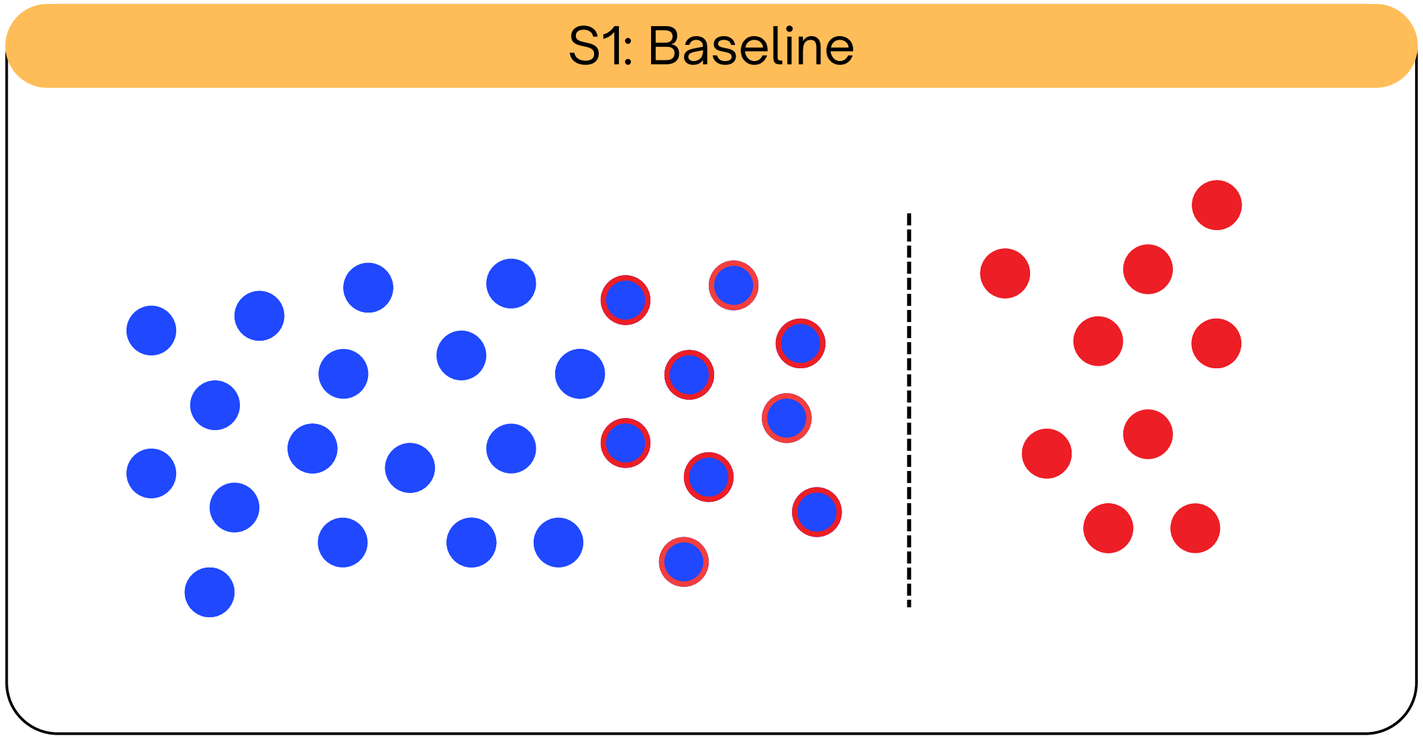
They learn to gradually remove noise, generating synthetic data from random noise.

(Training only)

DIR: Disparate Impact Remover
LFR: Learning Fair Representations

STUDY DESIGN

Overview of the Six Experimental



- No rebalancing or synthetic data

STUDY DESIGN Overview of the Six Experimental

S2: Fair pre-processing



Apply reweighting, DIR or LFR to mitigate bias in training data.

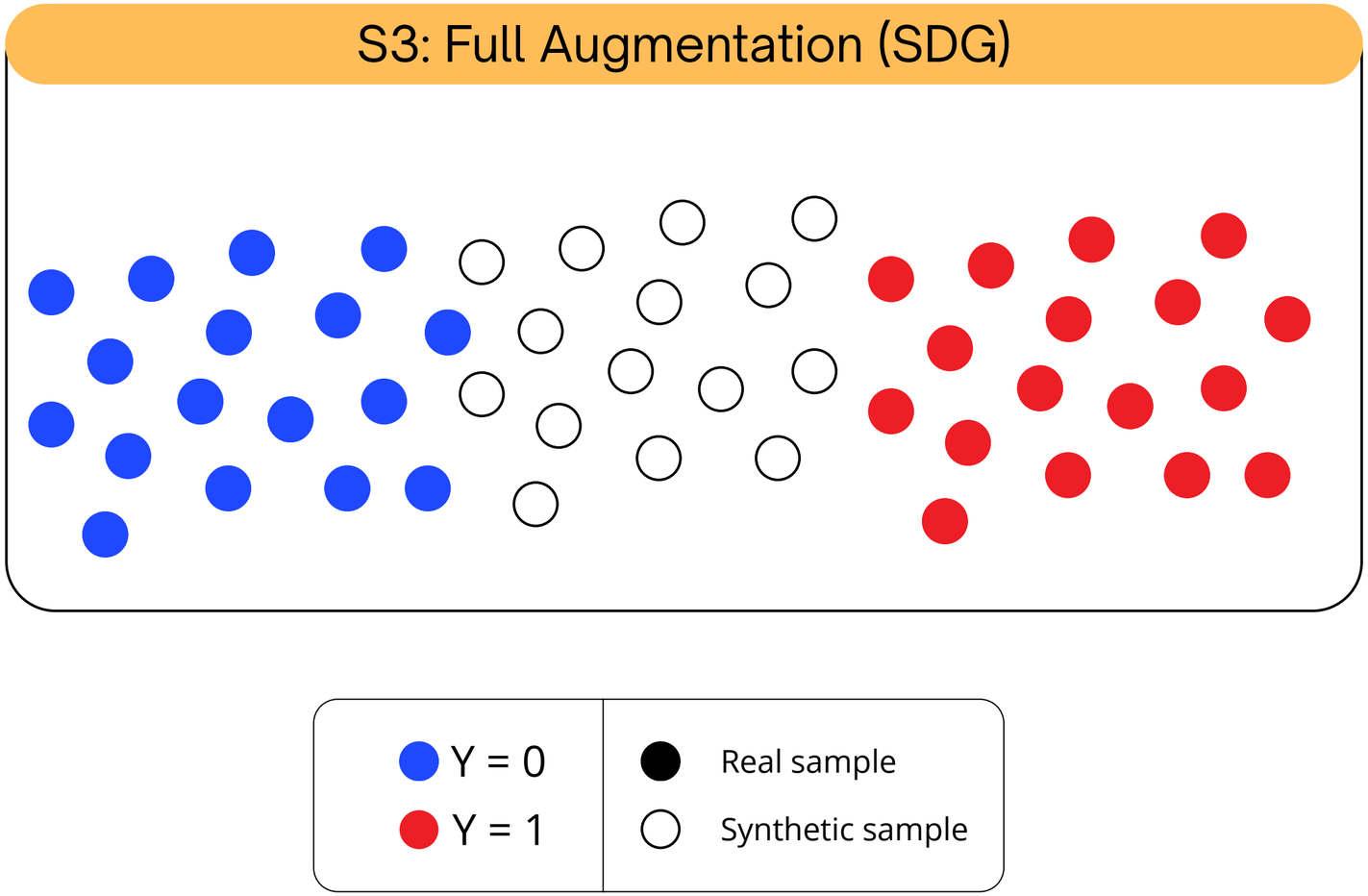
Y = 0
Y = 1

- No synthetic data generation

S: Scenario
Y: Target variable
DIR: Disparate Impact Remover
LFR: Learning Fair Representations

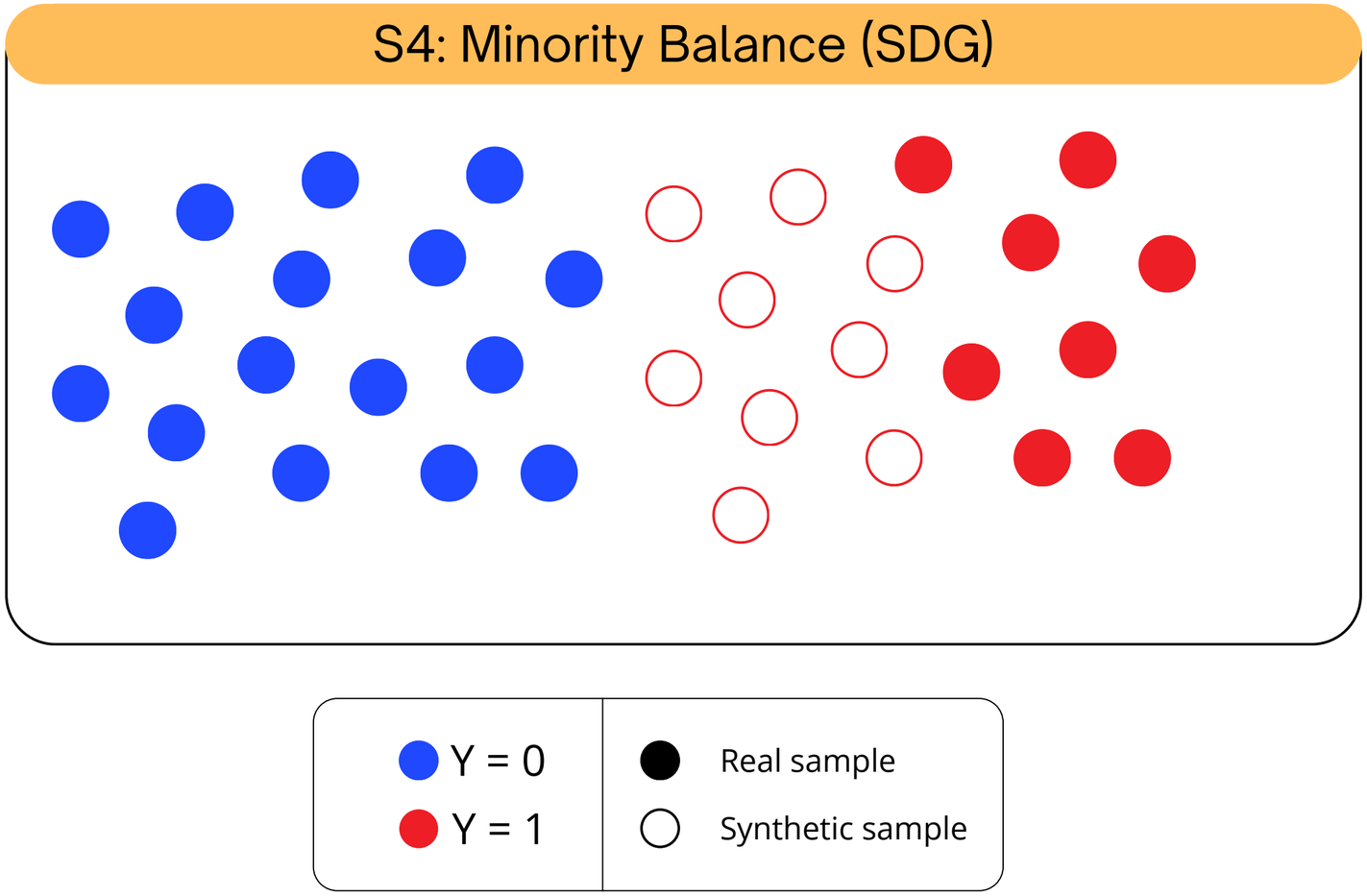
STUDY DESIGN

Overview of the Six Experimental



Generate synthetic data for the entire dataset (α = 1.5, 2, 3)

STUDY DESIGN Overview of the Six Experimental

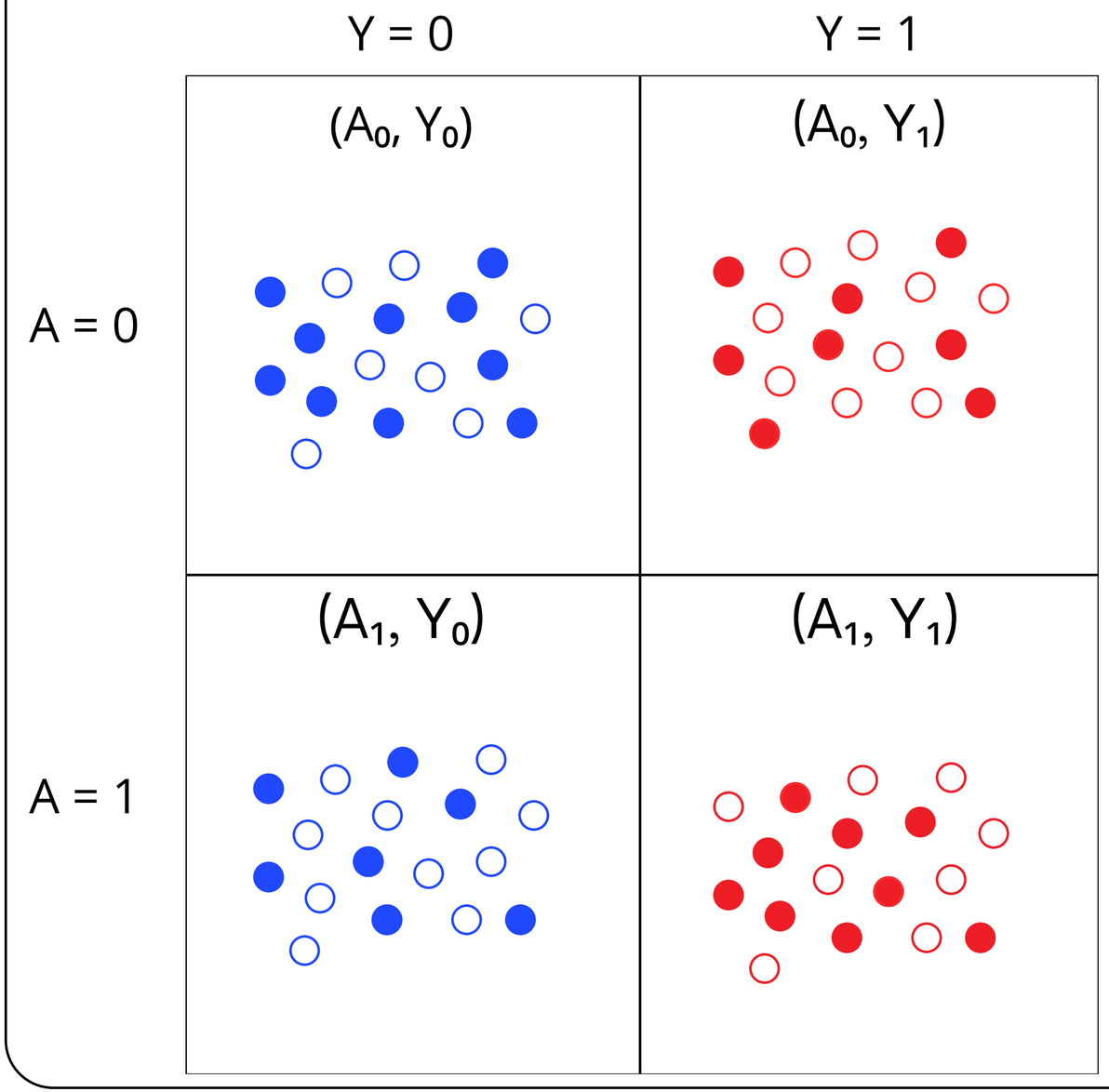


- Generate synthetic data only for the minority class.
- N(Y = 0) > N(Y = 1)

STUDY DESIGN

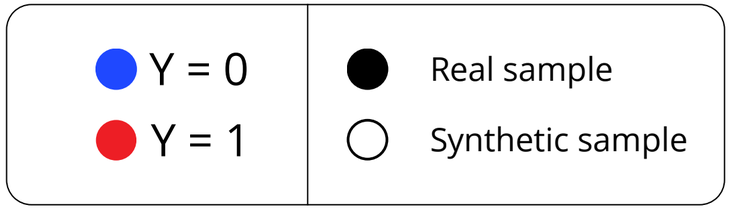
Overview of the Six Experimental

S5: Subgroup (A × Y)



- Generate synthetic data for each combination of protected attribute (A) and target (Y)

- $\forall\ a \in \{0,1\}, \quad (A=a,Y=0) \neq (A=a,Y=1)$



STUDY DESIGN Overview of the Six Experimental

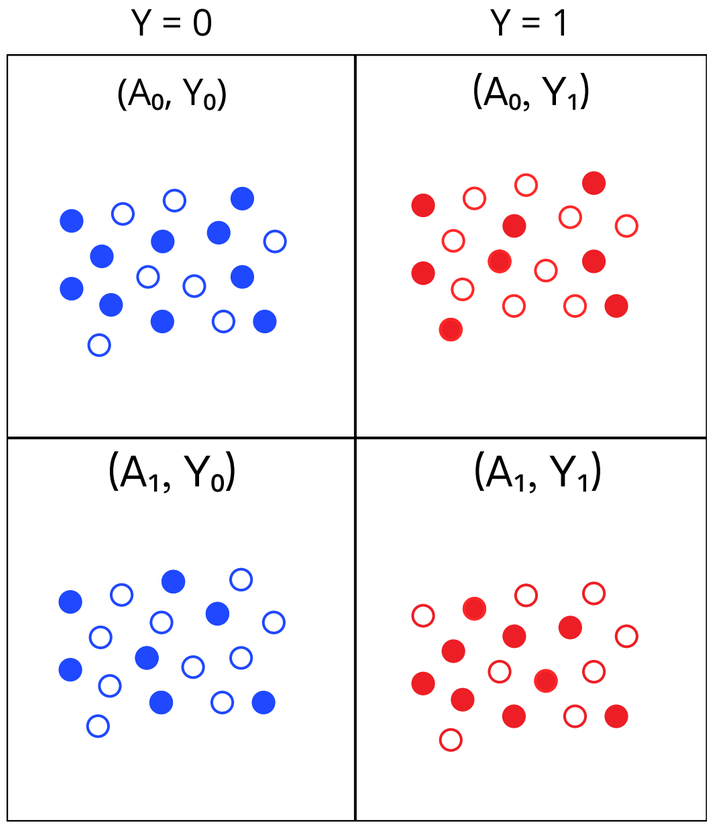
S6: Reweighting + S5



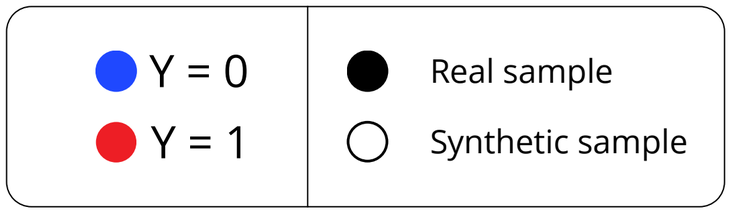
Reweighting

+

S5



- Apply reweighting and then generate subgroup-conditioned synthetic data



STUDY DESIGN

A systematic, factorial evaluation

≈ 840 runs per dataset

3 Datasets

4 Generators

3 Mitigators
only S2

3 Scalar α
only S3

3 Classifiers

6 Scenarios

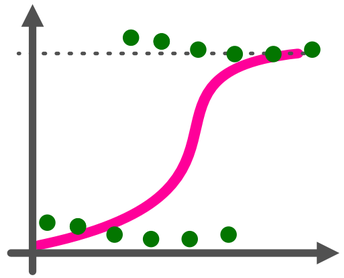
10 Seeds

These combinations resulted in approximately 840 runs per dataset.

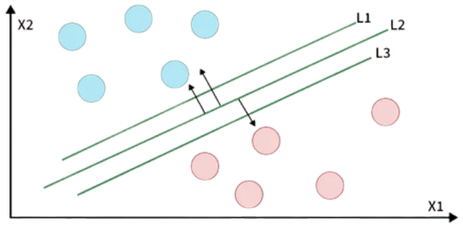
STUDY DESIGN

Predictive Models

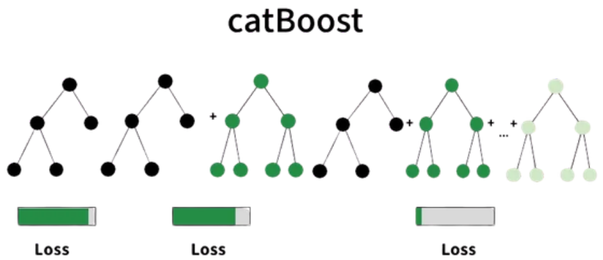
Model Training



Logistic Regression



SVM

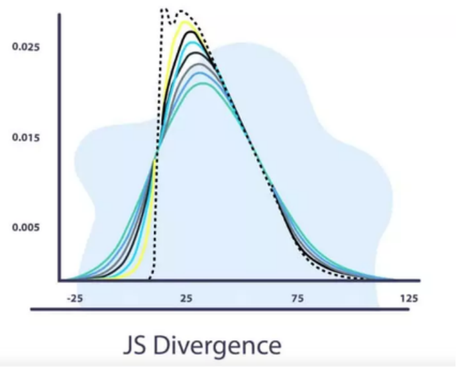


CatBoost

- Train and evaluate predictive models on datasets from each scenario
- Each experiment is repeated with 10 random seeds

STUDY DESIGN Evaluation

Evaluation

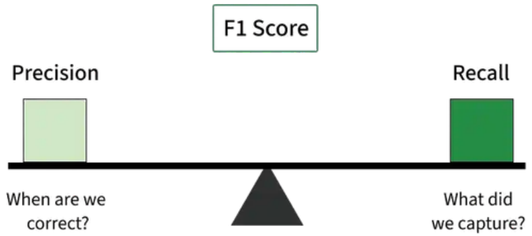


(Data Fidelity)

Jensen-Shanon Divergence

- JS Divergence between real and synthetic data

JSD = 0
Faithful



(Utility)

F1-score (macro-average)

(Fairness)



- Disparate Impact (DI)
- Equal Opportunity Difference (EOD)
- Average Absolute Odds Difference (AAOD)

$$\mathrm{DI} = \frac{P(\hat{Y} = 1 \mid A = 0)}{P(\hat{Y} = 1 \mid A = 1)}$$

$Underprivileged$

$Privileged$

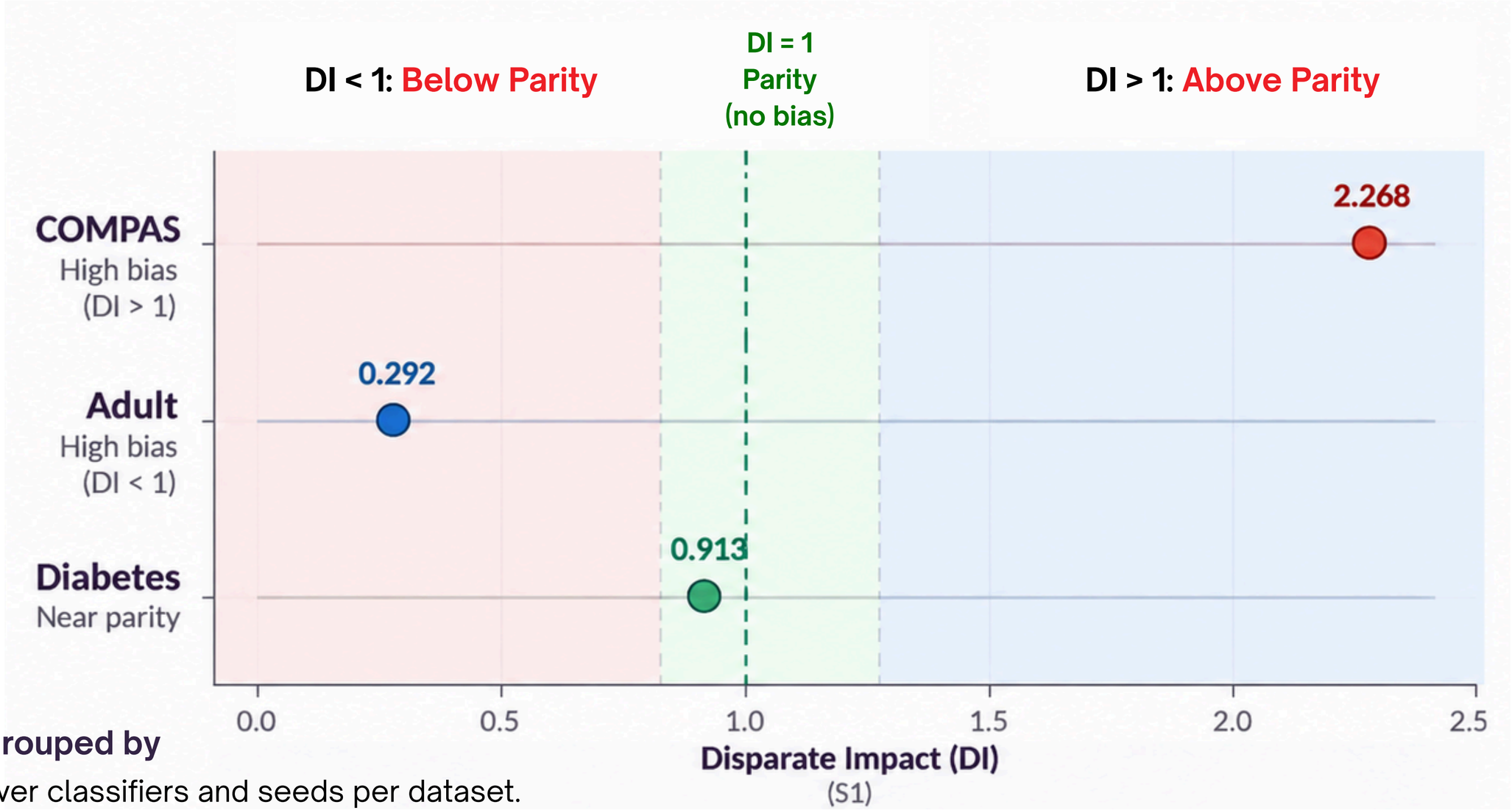
DI = 1
Parity (no bias)

Post-inference

EOD and AOD = 0
Parity (no bias)

SYNTHESIS

Initial bias determines the right strategy



Results grouped by

Mean DI over classifiers and seeds per dataset.

RESULT 1: COMPAS

More synthetic data, more disparity

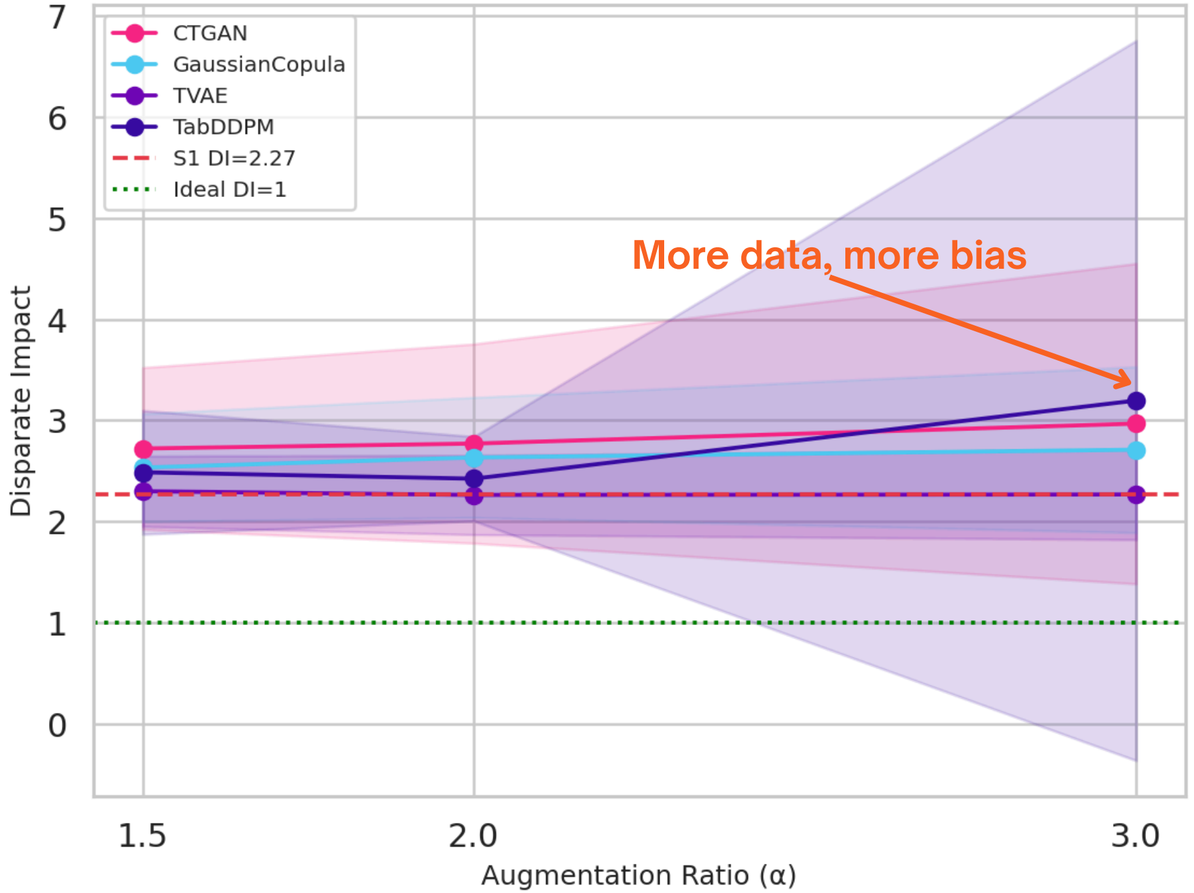
Baseline (S1): **DI 2.268**, F1 0.631

S3: full-data augmentation

	F1	DI	ΔDI
α = 1.5	0.623	2.511	+0.243
α = 2.0	0.614	2.523	+0.255
α = 3.0	0.608	2.786	+0.518

$$\frac{P(\text{recidivism} \mid \text{African American})}{P(\text{recidivism} \mid \text{Caucasian})}$$

0.537 / 0.242 = 2.22



Results grouped by

Disparate Impact vs augmentation ratio, per generator and α (mean over classifiers and seeds). Red dashed line: baseline DI = 2.27. Green dotted line: parity

RESULT 2: COMPAS

Targeting changes the outcome

Scenario	F1	DI	ΔDI	AAOD
S1: Baseline	0.631	2.268	-	0.331
S2: Reweighing	0.618	1.225	**−1.042**	0.099
S2: LFR	0.481	1.257	**−1.010**	0.086
S3: α = 3.0	0.608	2.786	**+0.518**	0.341
S4: Minority oversampling	0.638	1.326	**−0.941**	0.141
S5: Subgroup-conditioned	0.648	1.397	**−0.871**	0.164
S6: Hybrid	0.630	1.301	**−0.967**	0.123

Cost of LFR

Best trade-off

- S5 reaches F1 0.648 above the untouched baseline while cutting DI by 0.87.
- S6 matches baseline utility exactly (0.630) with the largest DI reduction of any synthetic strategy.
- Near-parity DI, but F1 collapses to 0.481 with ±0.182 spread significant degradation on every dataset.

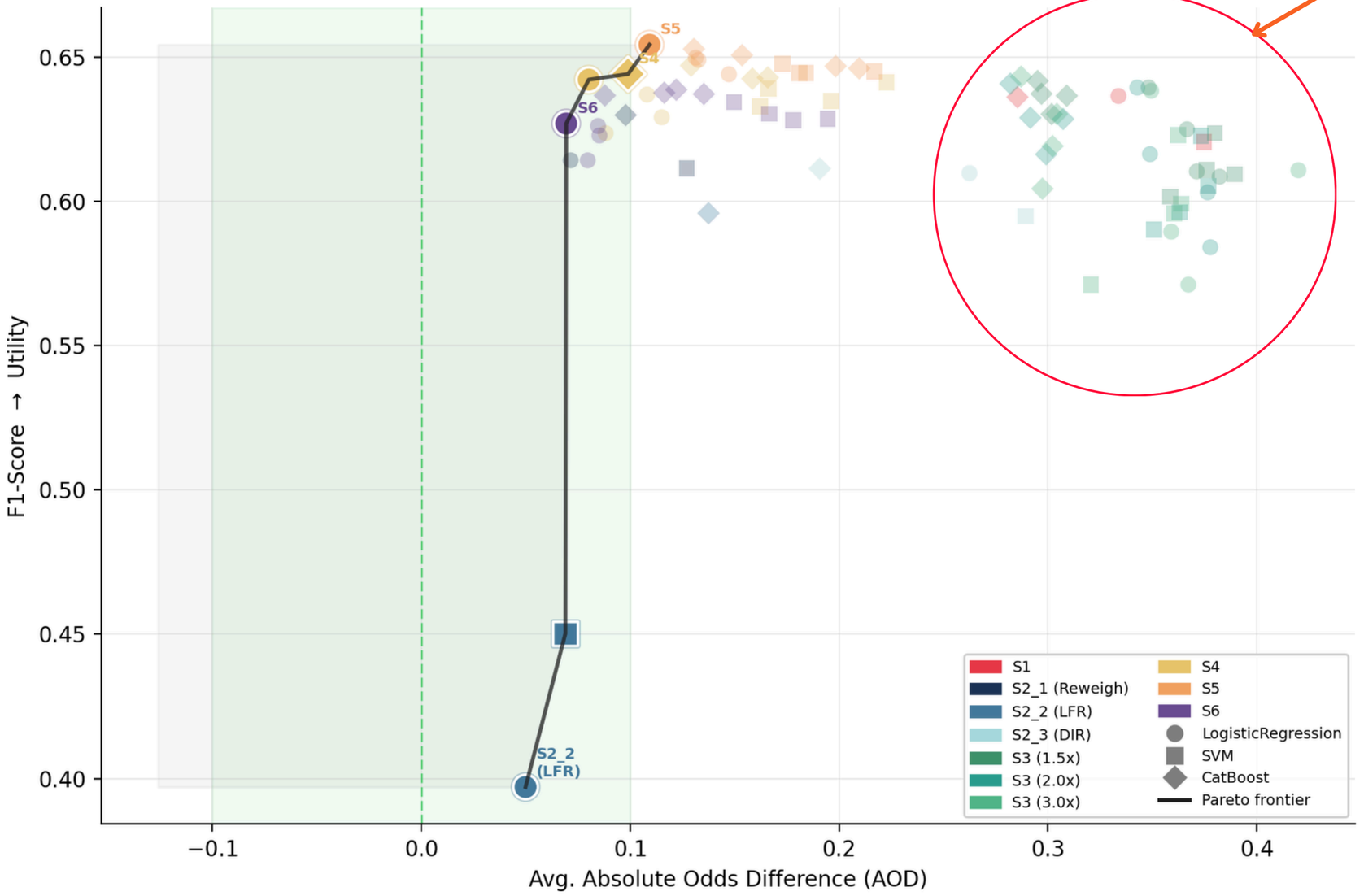
Statistical significance: S4 and S5 show significant improvements in DI (**$p < 0.05$**). S3 shows no significant effects on any dataset at any significance level (α). LFR causes significant reductions in F1 (**$p < 0.01$**).

Results grouped by

Mean over classifiers and seeds (n = 30 for S1–S2), and also over generators for S3–S6 (n = 120). ΔDI = DI − DI(S1).

RESULT 2: COMPAS

The utility-fairness frontier



S3 sits far right

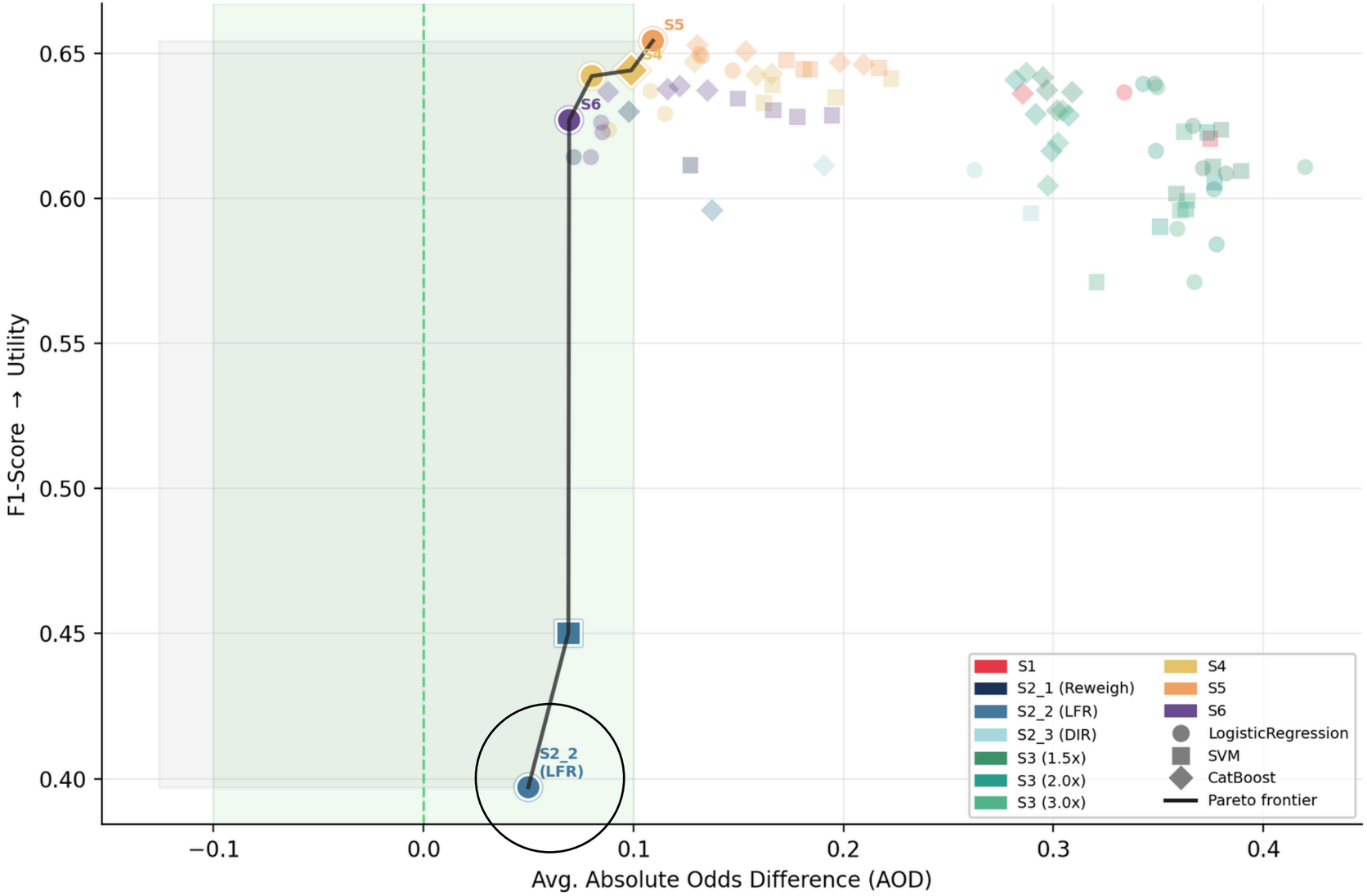
The worst fairness is observed in the AAOD range of 0.28-0.42.

Results grouped by

Each point is one configuration (scenario × generator × classifier), averaged over 10 seeds.

RESULT 2: COMPAS

The utility-fairness frontier

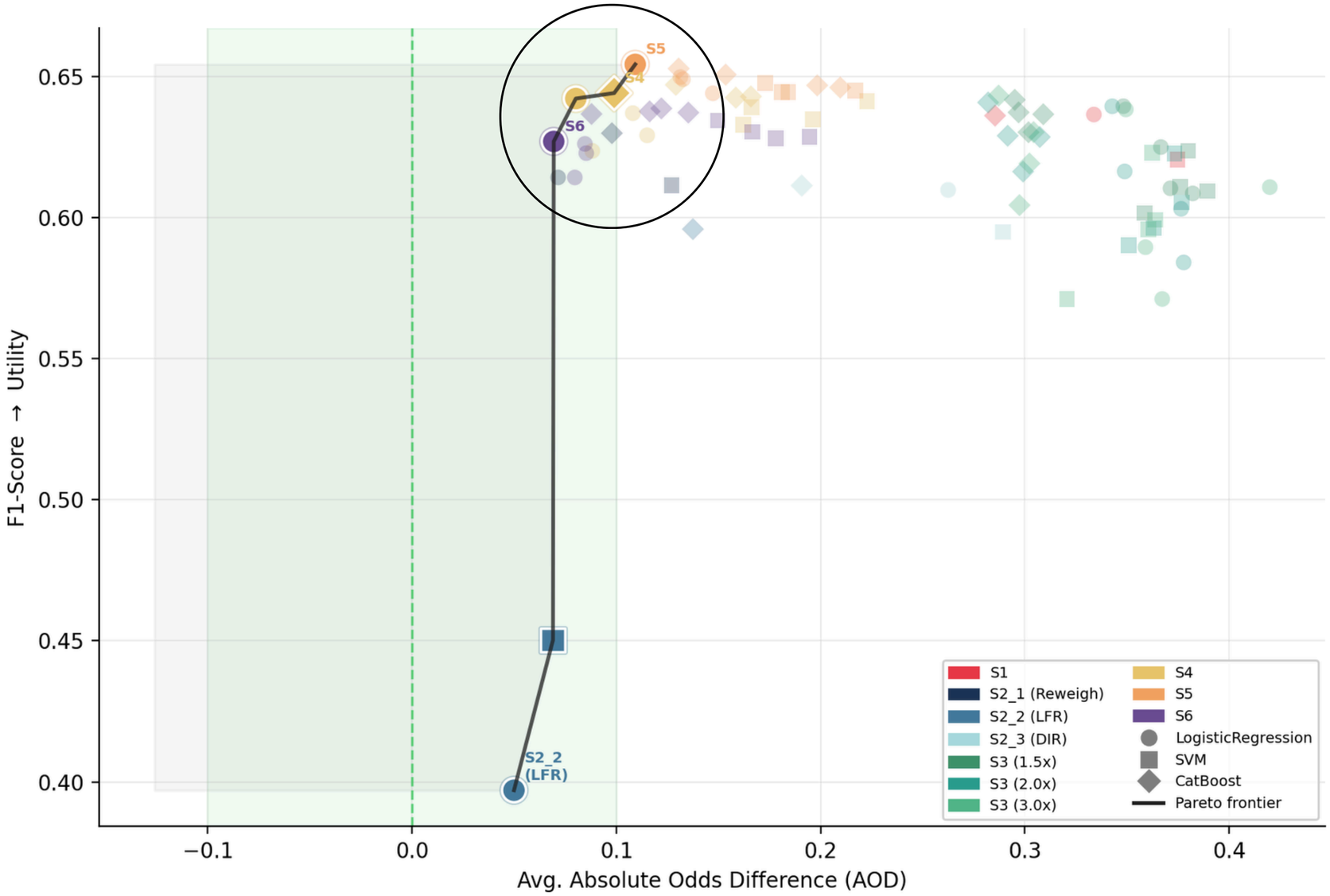


LFR sits bottom left

Excellent AAOD, but at F1 0.40-0.45. It buys fairness with utility the deployment cannot afford.

RESULT 2: COMPAS

The utility-fairness frontier



S5 and S6 define the frontier

S6 anchors the knee of the curve; S5 occupies the high-utility end.

RESULT 3: ADULT INCOME

Gender bias: a smaller gap, a harder one

Baseline DI 0.292, women receive the favourable prediction at about a third of the male rate.

Results grouped by

Mean over classifiers and seeds (n = 30 for S1-S2), and also over generators for S3-S6 (n = 120). ΔDI = DI − DI(S1).

Scenario	F1	DI	ΔDI	AAOD	EOD
S1 (Baseline)	0.655 ± 0.032	0.292 ± 0.024	0.000	0.118 ± 0.036	-0.116 ± 0.040
S2 Pre-processing					
DIRemover	0.626 ± 0.013	0.208 ± 0.068	-0.084	0.221 ± 0.110	-0.219 ± 0.113
LFR	0.305 ± 0.025	0.386 ± 0.087	+0.094	0.032 ± 0.020	-0.000 ± 0.038
Reweighing	0.631 ± 0.035	0.619 ± 0.062	+0.326	0.142 ± 0.038	0.142 ± 0.038
S3 Synthetic					
$\alpha = 1.5$	0.627 ± 0.047	0.273 ± 0.031	-0.020	0.140 ± 0.040	-0.138 ± 0.044
$\alpha = 2.0$	0.613 ± 0.058	0.266 ± 0.039	-0.026	0.147 ± 0.044	-0.146 ± 0.047
$\alpha = 3.0$	0.593 ± 0.082	0.257 ± 0.048	-0.035	0.148 ± 0.053	-0.147 ± 0.055
Synthetic Conditional					
S4	0.639 ± 0.040	0.371 ± 0.098	+0.079	0.133 ± 0.064	-0.090 ± 0.094
S5	0.648 ± 0.034	0.450 ± 0.134	+0.158	0.089 ± 0.037	-0.014 ± 0.093
S6	0.654 ± 0.033	0.396 ± 0.102	+0.104	0.099 ± 0.041	-0.054 ± 0.084

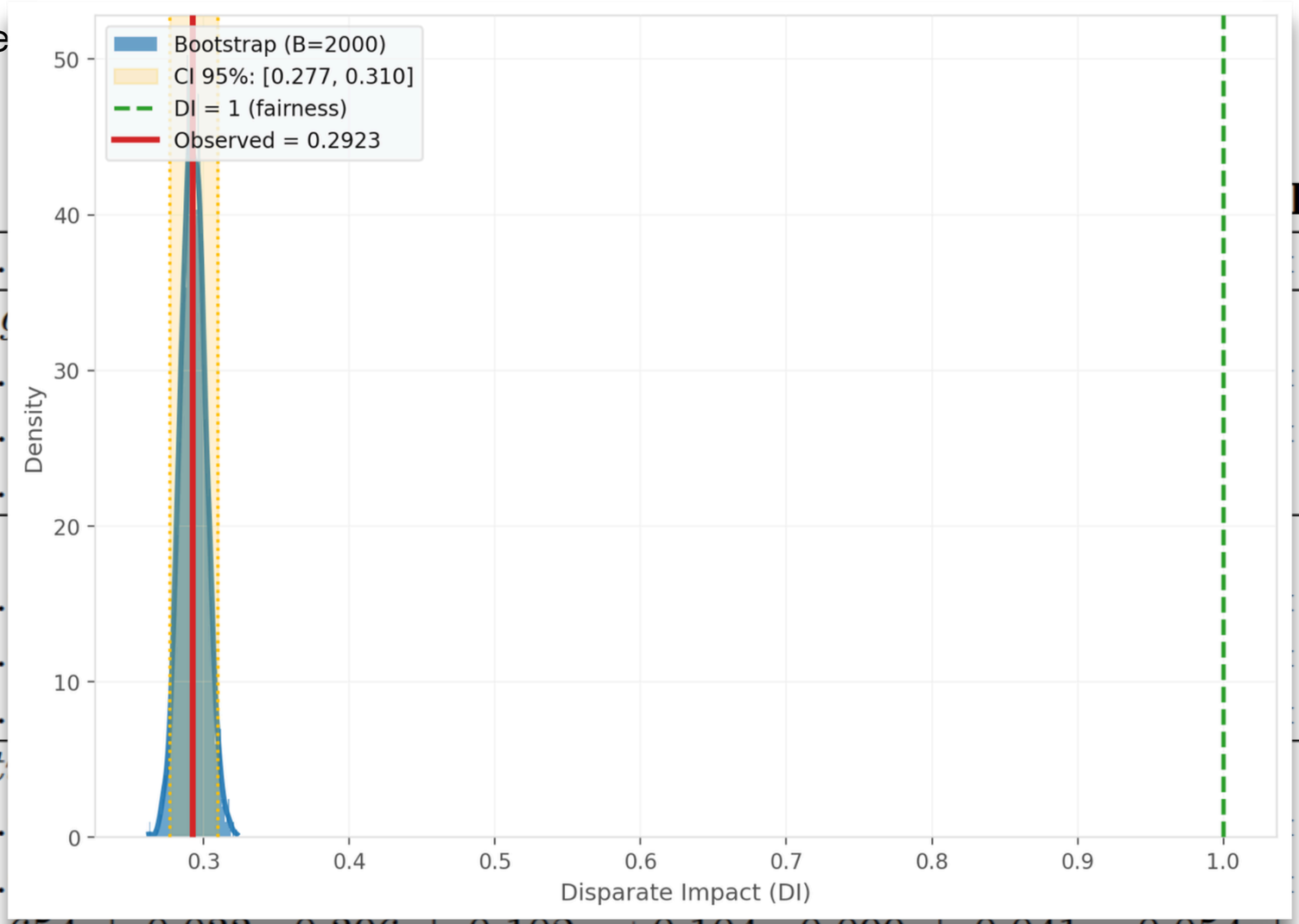
RESULT 3: ADULT INCOME

$$\frac{P(\text{income} > 50K \mid \text{female})}{P(\text{income} > 50K \mid \text{male})}$$

= 0.072 / 0.246 = 0.29

Gender bias: a smaller gap, a harder one

Baseline DI 0.292, women receive
third of the male rate.



Scenario					D
S1 (Baseline)	0.				0.040
S2 Pre-processing					
DIRemover	0.				0.113
LFR	0.				0.038
Reweighing	0.				0.038
S3 Synthetic					
$\alpha = 1.5$	0.				0.044
$\alpha = 2.0$	0.				0.047
$\alpha = 3.0$	0.				0.055
Synthetic Condit					
S4	0.				0.094
S5	0.				0.093
S6	0.654 ± 0.033	0.396 ± 0.102	$+0.104$	0.099 ± 0.041	$-0.054 \pm$ 0.084

RESULT 3: ADULT INCOME

Gender bias: a smaller gap, a harder one

S3 again drifts the wrong way: DI 0.292 to **0.257** at α = 3, with F1 falling from 0.655 to **0.593**.

Statistical significance: S4 and S5 show significant improvements in DI **($p < 0.05$)**. S3 shows no significant effects on any dataset at any significance level (α). LFR causes significant reductions in F1 **($p < 0.01$)**.

Scenario	F1	DI	ΔDI	AAOD	EOD
S1 (Baseline)	0.655 ± 0.032	0.292 ± 0.024	0.000	0.118 ± 0.036	-0.116 ± 0.040
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DIRemover	0.626 ± 0.013	0.208 ± 0.068	-0.084	0.221 ± 0.110	-0.219 ± 0.113
LFR	0.305 ± 0.025	0.386 ± 0.087	+0.094	0.032 ± 0.020	-0.000 ± 0.038
Reweighing	0.631 ± 0.035	0.619 ± 0.062	+0.326	0.142 ± 0.038	0.142 ± 0.038
S3 Synthetic					
$\alpha = 1.5$	0.627 ± 0.047	0.273 ± 0.031	-0.020	0.140 ± 0.040	-0.138 ± 0.044
$\alpha = 2.0$	0.613 ± 0.058	0.266 ± 0.039	-0.026	0.147 ± 0.044	-0.146 ± 0.047
$\alpha = 3.0$	0.593 ± 0.082	0.257 ± 0.048	-0.035	0.148 ± 0.053	-0.147 ± 0.055
Synthetic Conditional					
S4	0.639 ± 0.040	0.371 ± 0.098	+0.079	0.133 ± 0.064	-0.090 ± 0.094
S5	0.648 ± 0.034	0.450 ± 0.134	+0.158	0.089 ± 0.037	-0.014 ± 0.093
S6	0.654 ± 0.033	0.396 ± 0.102	+0.104	0.099 ± 0.041	-0.054 ± 0.084

RESULT 3: ADULT INCOME

Gender bias: a smaller gap, a harder one

Reweighing gives the largest fairness gain (DI 0.619) but costs utility (F1 0.631)

Scenario	F1	DI	ΔDI	AAOD	EOD
S1 (Baseline)	0.655 ± 0.032	0.292 ± 0.024	0.000	0.118 ± 0.036	-0.116 ± 0.040
S2 Pre-processing					
DIRemover	0.626 ± 0.013	0.208 ± 0.068	-0.084	0.221 ± 0.110	-0.219 ± 0.113
LFR	0.305 ± 0.025	0.386 ± 0.087	+0.094	0.032 ± 0.020	-0.000 ± 0.038
Reweighing	0.631 ± 0.035	0.619 ± 0.062	+0.326	0.142 ± 0.038	0.142 ± 0.038
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$\alpha = 3.0$	0.593 ± 0.082	0.257 ± 0.048	-0.035	0.148 ± 0.053	-0.147 ± 0.055
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S4	0.639 ± 0.040	0.371 ± 0.098	+0.079	0.133 ± 0.064	-0.090 ± 0.094
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S6	0.654 ± 0.033	0.396 ± 0.102	+0.104	0.099 ± 0.041	-0.054 ± 0.084

RESULT 3: ADULT INCOME

Gender bias: a smaller gap, a harder one

S5 lifts DI to **0.450** and nearly closes the opportunity gap: EOD −0.116 to **−0.014**, AAOD 0.118 to **0.089**.

Scenario	F1	DI	ΔDI	AAOD	EOD
S1 (Baseline)	0.655 ± 0.032	0.292 ± 0.024	0.000	0.118 ± 0.036	-0.116 ± 0.040
S2 Pre-processing					
DIRemover	0.626 ± 0.013	0.208 ± 0.068	-0.084	0.221 ± 0.110	-0.219 ± 0.113
LFR	0.305 ± 0.025	0.386 ± 0.087	+0.094	0.032 ± 0.020	-0.000 ± 0.038
Reweighing	0.631 ± 0.035	0.619 ± 0.062	+0.326	0.142 ± 0.038	0.142 ± 0.038
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$\alpha = 1.5$	0.627 ± 0.047	0.273 ± 0.031	-0.020	0.140 ± 0.040	-0.138 ± 0.044
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S6	0.654 ± 0.033	0.396 ± 0.102	+0.104	0.099 ± 0.041	-0.054 ± 0.084

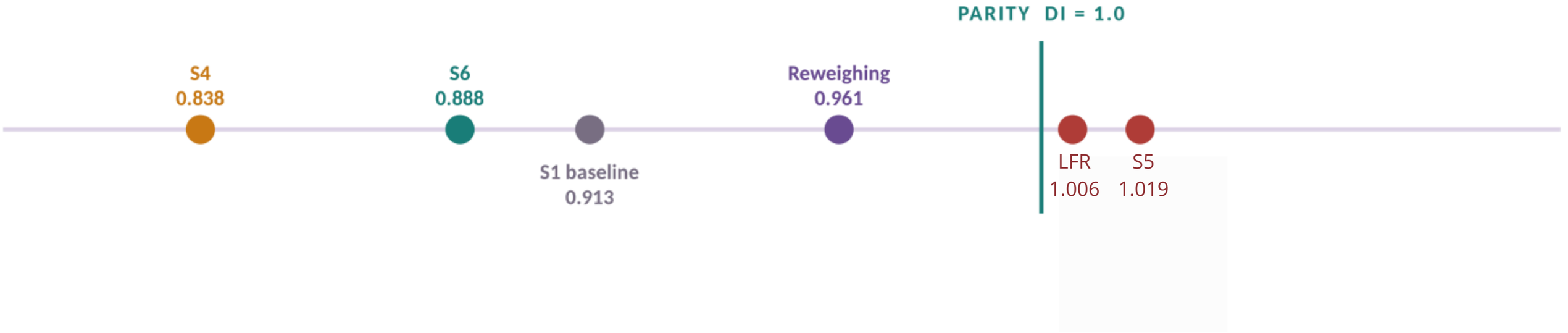
Best trade-off

RESULT 4: DIABETES

When there is little bias left to fix

Results grouped by

Mean over classifiers and seeds (n = 30 for S1-S2), and also over generators for S3-S6 (n = 120).

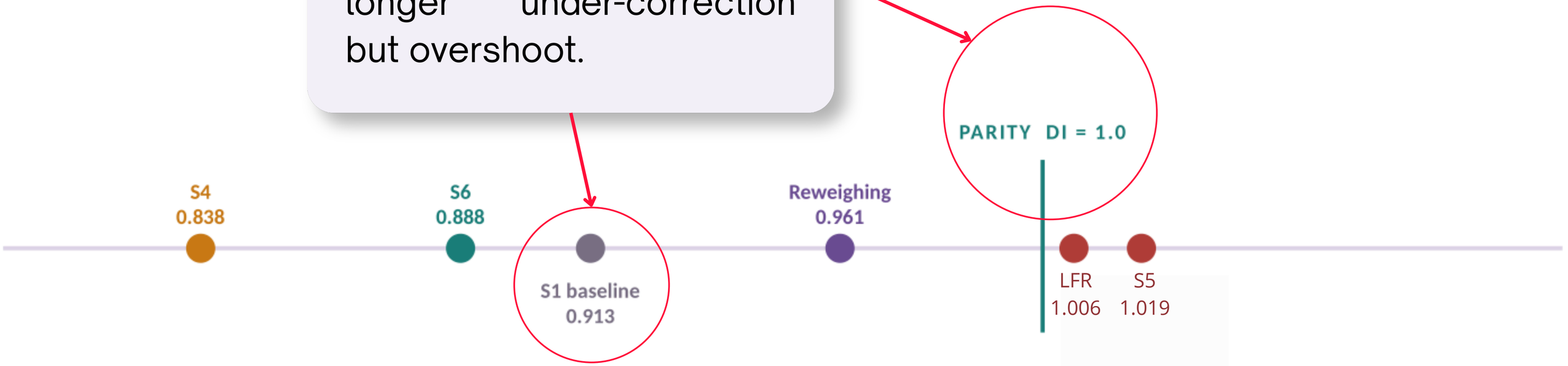


Statistical significance: S3 shows no significant effects on any dataset at any significance level (α). LFR causes significant reductions in F1 **($p < 0.01$)**.

RESULT 4: DIABETES

When there i

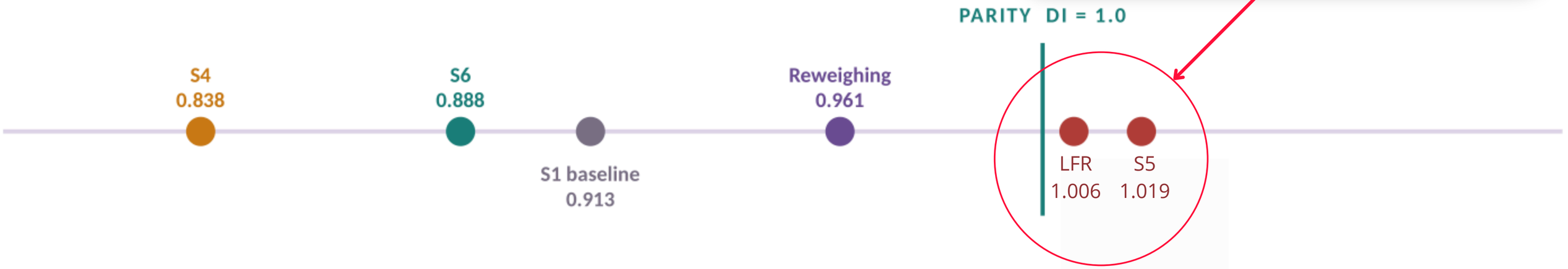
Baseline DI 0.913, close to parity, so the risk is no longer under-correction but overshoot.



RESULT 4: DIABETES

When there is little bias left to fix

Aggressive interventions cross the line: LFR reaches 1.006 and S5 reaches 1.019 the disparity is reversed, not removed.



CONCLUSIONS

1.Fidelity is not fairness

All generators had JSD < 0.15, faithfully reproducing the bias.

2.Never augment blindly

S3 amplified disparity on COMPAS and Adult.

3.Condition on the strata

Generating per (A × Y) cell improved fairness and utility.

4.Hybrid is the best default

S6 achieved the best fairness-utility balance across all datasets.

5.Calibrate to the starting bias

High-bias data rewards strong intervention; near-parity data punishes it with overshoot.

SCOPE

Limitations and what comes next

Limitations

- Binary protected attributes; no intersectionality.
- Adult and Diabetes subsampled to 10,000 rows; full-scale scalability untested.
- Binary classification only; regression fairness out of scope.
- Default hyperparameters to isolate data-centric effects.

SCOPE

Limitations and what comes next

Future work

- Extend the comparison to in-processing and post-processing stages.
- Multiple, intersecting protected attributes.
- A broader range of classifiers and generative architectures.

THANK YOU!

Questions & Discussion

Code & Experiments

Acknowledgments & Funding

CAPES - FINANCE CODE 001 | FAPERGS [22/2551-0000390-7] | CNPQ [308075/2021-8]

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RESULT: FIDELITY

The synthetic data is faithful and that is the problem

JSD < 0.15

$$0 \leq \mathrm{JSD} \leq \log_b(n)$$

for all four generators across most features, on all three datasets.

	Compas	Adult	Diabetes	
TVAE	0.008	0.007	0.020	lowest divergence overall
CTGAN	0.011	0.009	0.010	consistently low
Gaussian Copula	0.0116	0.025	0.049	higher on Adult, Diabetes
TabDDPM	0.044	0.0122	0.043	most variable

Results grouped by

Mean JSD between real and synthetic data (S3, α = 2.0), averaged over 10 seeds and all columns; lower = closer to real.